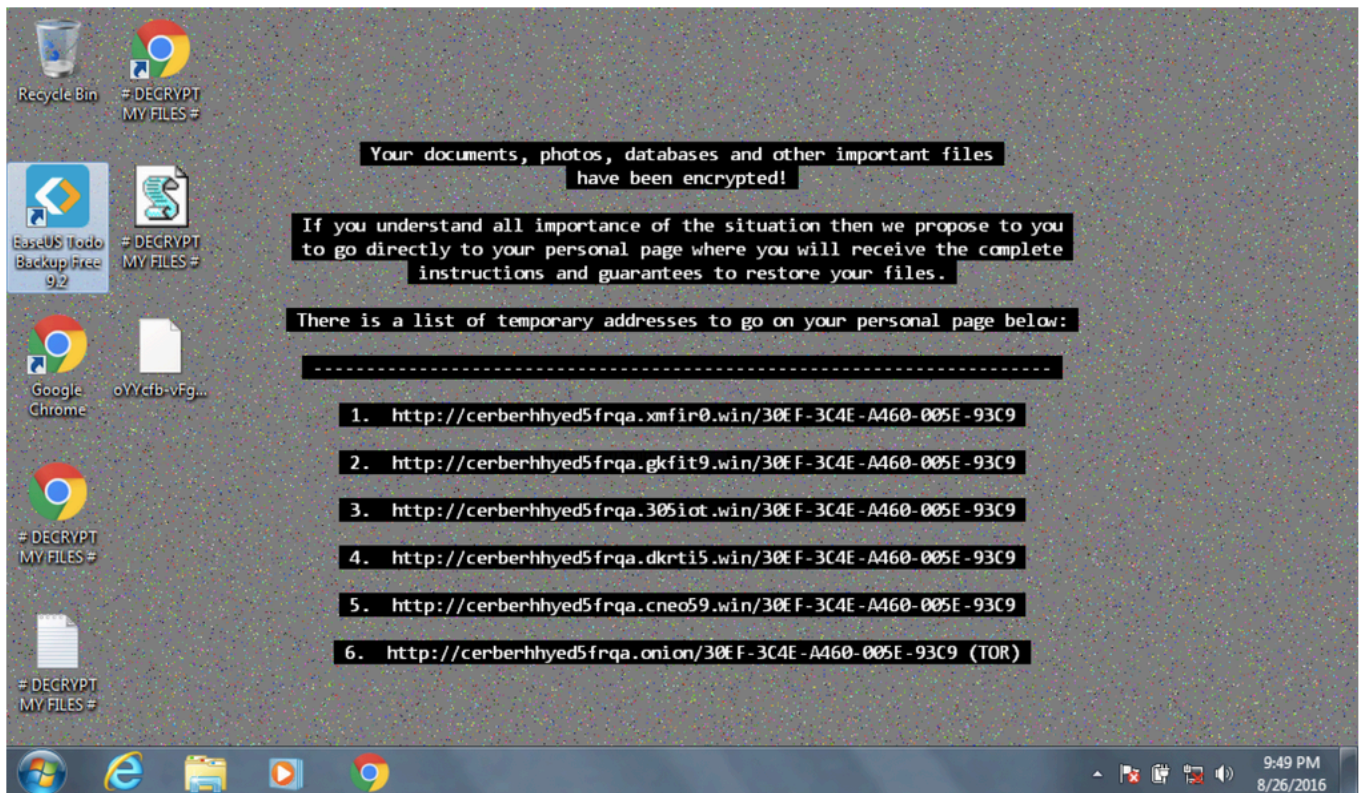


A cybersteps project

Date: 16. January 2026

Participants: Gregor

Project - Security Monitoring - Boss of the SOC



Project summary:

The goal is to investigate a ransomware-attack with the help of splunk. The hostname of the targeted machine is given: we8105desk

1. Finding the IP address

The first goal is to find the IP of the targeted system. I know, that the hostname is we8105desk. Searching the events for this hostname results in 116,823 events:

```
index=botstv1 we8105desk
```

Searching for the hostname in XmlWinEventLogs and looking at the 'src_ip's reveals that the vast majority events have the ip 192.168.250.100 as a source.

```
index=botsv1 we8105desk sourcetype=XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
```

src_ip

6 Werte, 50,262 % der Ereignisse

Ausgewählt

Berichte

Oberste Werte Spitzenwerte bis zum Zeitpunkt Seltene Werte

Ereignisse mit diesem Feld

Werte	Anzahl	%
192.168.250.100	52.270	99,649 %
192.168.250.255	69	0,132 %
127.0.0.1	66	0,126 %
0.0.0.0	42	0,08 %
224.0.0.252	6	0,011 %
192.168.250.70	1	0,002 %

The IP address of we8105desk is 192.168.250.100.

2. Identifying Removable Media

The assumption is, that the system got infected by the ransomware via an USB-device.

This search shows the name of the USB-device, which was used by we8105desk :

```
index=botsv1 sourcetype=winregistry friendlyname | table host object data
```

we8105desk	friendlyname	MIRANDA_PRI
we8105desk	friendlyname	MIRANDA_PRI

USB-Name: MIRANDA_PRI

3. Identifying the Malicious File

```
index=botsv1 sourcetype=XmlWinEventLog:Microsoft-Windows-Sysmon/Operational  
host=we8105desk (CommandLine="*d:\\*" OR ParentCommandLine="*d:\\*") | table  
_time CommandLine ParentCommandLine | sort _time
```

The file Miranda_Tate_unveiled.dotm was triggered by Microsoft Word and it was located on the D:-drive.

To reveal suspicious processes I used the following search:

This revealed a suspicious process:

4490

The next step was to identify the IP address of the file server, which we8105desk was connected during the ransomware outbreak.

```
index=botsv1 sourcetype=XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
host=we8105desk src=we8105desk.waynecorpinc.local
```

EventDescription 1 Wert, 100 % der Ereignisse Ausgewählt Ja Nein		
Berichte Oberste Werte Spitzenwerte bis zum Zeitpunkt Seltene Werte Ereignisse mit diesem Feld		
Werte	Anzahl	%
Network Connect	52.378	100 %

Using the stats command reveals two IP addresses, which are responsible for the majority of connections:

```
index=botsv1 sourcetype=XmlWinEventLog:Microsoft-Windows-Sysmon/Operational  
host=we8105desk src=we8105desk.waynecorpinc.local | stats count by dest_ip |  
sort - count
```

Ereignisse		Muster	Statistik (10.000)	Visualisierung
20 pro Seite		Format	Vorschau	
dest_ip				count
192.168.250.20				1537
192.168.2.50				1448
127.0.0.1				66
192.168.250.255				54
255.255.255.255				42
239.255.255.250				11

192.168.250.20 and 192.168.2.50 both have a large amount of logs associated with this host.

Searching in the Windows registry data shows key_path field entries that reference one of my top two internal IP addresses from the earlier search

```
index=botsv1 sourcetype=winregistry host=we8105desk fileshare
```

key_path

1 Wert, 100 % der Ereignisse

Ausgewählt

Ja

Nein

Berichte

Oberste Werte

Spitzenwerte bis zum Zeitpunkt

Seltene Werte

Ereignisse mit diesem Feld

Werte	Anzahl	%
HKU\s-1-5-21-67332772-3493699611-3403467266-1109\software\microsoft\windows\currentversion\explorer\mountpoint s2\##192.168.250.20#fileshare	818	100 %

Fileserver IP address: 192.168.250.20

Identifying the Hostname of the File Server

Searching for the IP-address also provides the hostname:

```
index=botsv1 sourcetype="XmlWinEventLog:Microsoft-Windows-  
Sysmon/Operational" 192.168.250.20
```

In DestinationHostname you can find the hostname, which is we9041srv.

DestinationHostname

>100 Werte, 38,8 % der Ereignisse

Ausgewählt

Ja

Nein

Berichte

Oberste Werte

Spitzenwerte bis zum Zeitpunkt

Seltene Werte

Ereignisse mit diesem Feld

10 oberste Werte	Anzahl	%
we9041srv	1.553	21,368 %
WE9041SRV	1.015	13,965 %
google-public-dns-a.google.com	382	5,256 %
we9041srv.waynecorpinc.local	44	0,605 %
r.arin.net	25	0,344 %
we9748srv.waynecorpinc.local	20	0,275 %
we9748srv	19	0,261 %
ns1.velcom.com	15	0,206 %
ns2.velcom.com	14	0,193 %
a184-26-161-192.deploy.static.akamaitechnologie s.com	13	0,179 %

6. Identifying the First Suspect Domain Visited by the Victim

The next step is to find out which suspect domain was visited from the targeted system. For that I looked up the DNS stream data and excluded unsuspect domains like microsoft.com etc.:

```
index=botsv1 sourcetype=stream:DNS src=192.168.250.100 record_type=A NOT
(query{ }=*.microsoft.com OR query{ }=*.waynecorpinc.local OR
query{ }=*.bing.com OR query{ }=isatap OR query{ }=wpad OR
query{ }=*.windows.com OR query{ }=*.msftncsi.com) | table _time query{ } src
dest | reverse
```

This revealed the following domain:

Ereignisse	Muster	Statistik (3)	Visualisierung
20 pro Seite	Format	Vorschau	
_time	query{ }	src	dest
2016-08-24 17:15:12.668	cerberhyedfrqa.xmfi0.win cerberhyedfrqa.xmfi0.win	192.168.250.100	192.168.250.20
2016-08-24 16:48:12.267	solidaritedproximate.org solidaritedproximate.org	192.168.250.100	192.168.250.20
2016-08-24 16:49:24.308	ipinfo.io ipinfo.io	192.168.250.100	192.168.250.20

Initial Connection to a suspicious domain: solidaritedeproximityte.org

7. Identifying Cryptor Code Filename and Origin

To find out the Filename of the ransomware I searched the http-stream with the targeted IP as source.

```
index=botsv1 sourcetype=http-stream src=192.168.250.100 url=/mhtr.jpg |  
table _time src dest http.hostname
```

20 pro Seite ▾					Format	Vorschau ▾
_time ▾		src ▾	✓	dest ▾	✓	http.hostname ▾
2016-08-24 16:48:14.620		192.168.250.100		92.222.104.182		92.222.104.182
2016-08-24 16:48:13.492		192.168.250.100		solidaritedeproxinite.org		solidaritedeproxinite.org

A suspicious file stood out, which was mhtr.jpg. A quick google search confirmed that this filename is related to the cerber-botnet.

Looking into the firewall logs (fgt_utm) show a lot of entries:

```
index=botsv1 sourcetype=fgt_utm src=192.168.250.100 app="Cerber.Botnet" |  
reverse
```

i	Uhrzeit	Ereignis
>	24.08.16 16:49:24,000	Aug 24 10:49:24 192.168.250.1 date=2016-08-24 time=10:49:24 devname=gotham-fortigate devid=FGT6004614044725 logid=1059028704 type=utm subtype=app-ctrl eventtype=app-ctrl-all level=information vd="root" appid=42196 user="" srcip=192.168.250.100 srcport=50720 srcintf="internal3" dstip=85.93.0.0 dstport=6892 proto=17 service="udp/6892" policyid=10 sessionid=4191786 applist="HoneyPot-Access" appcat="Botnet" app="Cerber.Botnet" action=pass crscore=50 crlevel=critical msg="Botnet: Cerber.Botnet," apprisk=critical host = 192.168.250.1 : source = udp:514 : sourcetype = fgt_utm
>	24.08.16 16:49:24,000	Aug 24 10:49:24 192.168.250.1 date=2016-08-24 time=10:49:24 devname=gotham-fortigate devid=FGT6004614044725 logid=1059028704 type=utm subtype=app-ctrl eventtype=app-ctrl-all level=information vd="root" appid=42196 user="" srcip=192.168.250.100 srcport=50720 srcintf="internal3" dstip=85.93.0.1 dstport=6892 proto=17 service="udp/6892" policyid=10 sessionid=4191787 applist="HoneyPot-Access" appcat="Botnet" app="Cerber.Botnet" action=pass crscore=50 crlevel=critical msg="Botnet: Cerber.Botnet," apprisk=critical host = 192.168.250.1 : source = udp:514 : sourcetype = fgt_utm
>	24.08.16 16:49:24,000	Aug 24 10:49:24 192.168.250.1 date=2016-08-24 time=10:49:24 devname=gotham-fortigate devid=FGT6004614044725 logid=1059028704 type=utm subtype=app-ctrl eventtype=app-ctrl-all level=information vd="root" appid=42196 user="" srcip=192.168.250.100 srcport=50720 srcintf="internal3" dstip=85.93.0.2 dstport=6892 proto=17 service="udp/6892" policyid=10 sessionid=4191788 applist="HoneyPot-Access" appcat="Botnet" app="Cerber.Botnet" action=pass crscore=50 crlevel=critical msg="Botnet: Cerber.Botnet," apprisk=critical host = 192.168.250.1 : source = udp:514 : sourcetype = fgt_utm
>	24.08.16 16:49:24,000	Aug 24 10:49:24 192.168.250.1 date=2016-08-24 time=10:49:24 devname=gotham-fortigate devid=FGT6004614044725 logid=1059028704 type=utm subtype=app-ctrl eventtype=app-ctrl-all level=information vd="root" appid=42196 user="" srcip=192.168.250.100 srcport=50720 srcintf="internal3" dstip=85.93.0.3 dstport=6892 proto=17 service="udp/6892" policyid=10 sessionid=4191789 applist="HoneyPot-Access" appcat="Botnet" app="Cerber.Botnet" action=pass crscore=50 crlevel=critical msg="Botnet: Cerber.Botnet," apprisk=critical host = 192.168.250.1 : source = udp:514 : sourcetype = fgt_utm
>	24.08.16 16:49:24,000	Aug 24 10:49:24 192.168.250.1 date=2016-08-24 time=10:49:24 devname=gotham-fortigate devid=FGT6004614044725 logid=1059028704 type=utm subtype=app-ctrl eventtype=app-ctrl-all level=information vd="root" appid=42196 user="" srcip=192.168.250.100 srcport=50720 srcintf="internal3" dstip=85.93.0.4 dstport=6892 proto=17 service="udp/6892" policyid=10 sessionid=4191790 applist="HoneyPot-Access" appcat="Botnet" app="Cerber.Botnet" action=pass crscore=50 crlevel=critical msg="Botnet: Cerber.Botnet," apprisk=critical host = 192.168.250.1 : source = udp:514 : sourcetype = fgt_utm
>	24.08.16 16:49:24,000	Aug 24 10:49:24 192.168.250.1 date=2016-08-24 time=10:49:24 devname=gotham-fortigate devid=FGT6004614044725 logid=1059028704 type=utm subtype=app-ctrl eventtype=app-ctrl-all level=information vd="root" appid=42196 user="" srcip=192.168.250.100 srcport=50720 srcintf="internal3" dstip=85.93.0.5 dstport=6892 proto=17 service="udp/6892" policyid=10 sessionid=4191791 applist="HoneyPot-Access" appcat="Botnet" app="Cerber.Botnet" action=pass crscore=50 crlevel=critical msg="Botnet: Cerber.Botnet," apprisk=critical host = 192.168.250.1 : source = udp:514 : sourcetype = fgt_utm

9. Determine Which Signatures Specific to the Ransomware Alerted

```
index=botsv1 sourcetype=suricata alert.signature=*cerber*
```

To specify the alerts we search for cerber-related alert-signature.

alert.signature

3 Werte, 100 % der Ereignisse

Ausgewählt

Ja

Nein

Berichte

Oberste Werte

Spitzenwerte bis zum Zeitpunkt

Seltene Werte

Ereignisse mit diesem Feld

Werte	Anzahl	%	
ETPRO TROJAN Ransomware/Cerber Checkin Error ICMP Response	2	40 %	
ETPRO TROJAN Ransomware/Cerber Onion Domain Lookup	2	40 %	
ETPRO TROJAN Ransomware/Cerber Checkin 2	1	20 %	

10. Damage Assessment - Identifying Encrypted Text Files

To assess the damage I looked for a way to see how much textfiles on we8105desk got encrypted.

```
index=botsv1 sourcetype=XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
host=we8105desk EventCode=2
TargetFilename="C:\\Users\\bob.smith.WAYNECORPINC\\*.txt" | stats
dc(TargetFilename)
```

This search revealed, that 406 textfiles were effected:

✓ 406 Ereignisse (24.08.16 00:00:00,000 bis 25.08.16 00:00:00,000)		Kein Abruf von Beispielergebnissen ▼	
Ereignisse	Muster	Statistik (1)	Visualisierung
20 pro Seite ▼	✎ Format	Vorschau ▼	
dc(TargetFilename) ↕			
406			

11. Damage Assessment - Identifying Distinct PDFs Encrypted

This search helped to determine the amount of encrypted PDF-files:

```
index=botsv1 sourcetype=*win* pdf dest=we9041srv.waynecorpinc.local
Source_Address=192.168.250.100 | stats dc(Relative_Target_Name)
```

This search revealed, that 257 PDF files were effected.

12. Identifying Redirection Post Encryption to a Domain

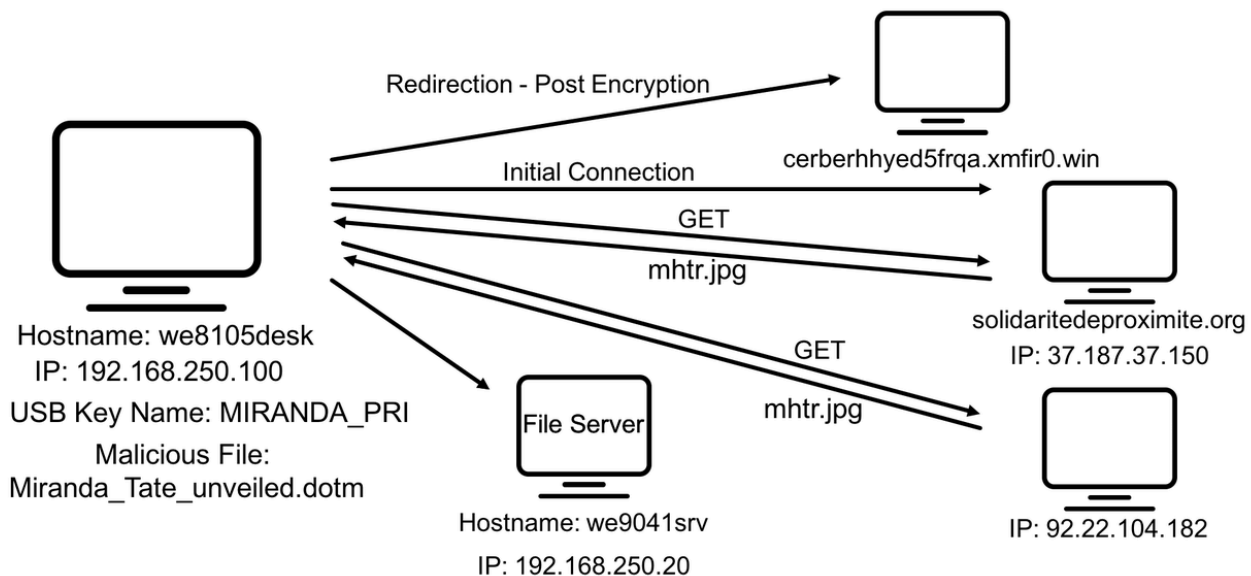
To find out any connection, which happned after the encryption I used the following search:

```
index=botsv1 sourcetype=stream:DNS src=192.168.250.100 record_type=A NOT
(query{ }=*.microsoft.com OR query{ }=*.waynecorpinc.local OR
query{ }=*.bing.com OR query{ }=isatap OR query{ }=wpad OR
query{ }=*.windows.com OR query{ }=*.msftncsi.com) | table _time query{ } src
dest
```

_time	query{ }	src	dest
2016-08-24 16:49:24.308	ipinfo.io ipinfo.io	192.168.250.100	192.168.250.20
2016-08-24 16:48:12.267	solidaritedeproximityte.org solidaritedeproximityte.org	192.168.250.100	192.168.250.20
2016-08-24 17:15:12.668	cerberhhyed5frqa.xmfir0.win cerberhhyed5frqa.xmfir0.win	192.168.250.100	192.168.250.20

The user got redirected to cerberhhyed5frqa.xmfir0.win after the encryption.

Summary of the ransomware-event



- User (or someone) inserted an infected USB drive into Bob Smith's workstation (not sure why!?!)
- Word document spawned a Suspicious Process which spawned additional processes
- File was downloaded after calls were made to a FQDN and IP address
- Encryption of files begin on both local disk and shares

- Redirection to an external site with notification of encryption occurred

Timeline of the events

